

Q.20 Explain the key physiochemical properties of milk, such as specific gravity, acidity, and freezing point. Discuss how these properties are used as quality control parameters to detect adulteration with water and ensure the freshness of milk.

No. of Printed Pages : 4

189223

Roll No.

Level 3, 2nd Sem / Food Processing

Subject : Milk Processing

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Very short answer Questions. Attempt all ten questions. (10x1=10)

- Q.1 What is the main protein found in milk?
- Q.2 Yellowish color of milk fat is primarily due to_____
- Q.3 What is the approximate freezing point of milk?
- Q.4 What is the standard method for determining the number of viable bacteria in a milk sample?
- Q.5 What process breaks down milk fat globules into smaller particles to prevent cream from separating?
- Q.6 What is the purpose of pasteurization in milk processing?

- Q.7 What is a common time and temperature combination for High-Temperature Short-Time (HTST) pasteurization?
- Q.8 What does MRL stand for in the context of chemical residues in food?
- Q.9 Which Indian body sets the quality standards for milk and milk products?
- Q.10 What is a key difference between organic and conventional milk regarding cow treatment?

SECTION-B

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

- Q.11 Describe the main factors affecting the composition of milk, highlighting how feeding regime influences milk fat and solids-not-fat (SNF) content.
- Q.12 How does the freezing point of milk help in determining its purity? Discuss the typical freezing point of pure milk and how adding water affects this value.
- Q.13 Why is it crucial to determine the microbial load in milk? Examine the sources of microbial contamination from the point of milking to processing.

(2)

189223

- Q.14 Differentiate between standardization and homogenization in milk processing.
- Q.15 Explain the objectives of pasteurization. Show how the effectiveness of pasteurization is monitored using an enzyme test.
- Q.16 Discuss the role of the Bureau of Indian Standards (BIS) in ensuring the quality of milk in India. Interpret the BIS standards for standardized milk regarding its fat and SNF content.
- Q.17 Discuss the common pathways through which milk and milk products can become contaminated with toxins.
- Q.18 Interpret the BIS standards for different classes of milk, such as Toned and Double Toned milk. Why are these specific standards for fat and SNF content necessary for consumer protection?

SECTION-C

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

- Q.19 Discuss the different types of pasteurization techniques used in the dairy industry. Describe the principles of High-Temperature Short-Time (HTST) pasteurization and explain its advantages over older methods.

(3)

189223